基于言语流畅性任务近红外光谱脑功能成像在抑郁障碍中的应用研究进展

范梦影 梁学军* 段惠峰 周林 胡伟 安佰海

联勤保障部队第九八八医院全军精神疾病防治研究所，河南 焦作 454000

[摘要] 抑郁障碍是一种高发病率和高自杀率的精神障碍，发病机制尚未明确，目前主要根据患者的症状及精神检查的结果进行诊断，缺乏客观的指标。近红外光谱脑功能成像（fNIRS）作为一种安全无创的脑功能成像技术，主要通过测量大脑皮层的血流动力学变化进行抑郁障碍的研究，言语流畅性任务（VFT）作为最常用的任务范式，能有效地评估抑郁障碍患者的言语能力和执行功能。本文综述了fNIRS结合VFT在抑郁障碍中的应用，探讨了其在诊断、鉴别及疗效评估中的价值，可能是抑郁障碍的潜在生物学标志物，并有望为患者提供个体化治疗。

[关键词] 抑郁障碍；近红外光谱脑功能成像；言语流畅性任务；诊断；疗效评估

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Research progress on the application of functional near-infrared spectroscopy based on speech fluency task in depressive disorders

FAN Mengying, LIANG Xuejun*, DUAN Huifeng, ZHOU Lin, HU Wei, AN Baihai

The 988th Hospital of the Joint Logistics Support Force, Institute of Mental Disease Prevention and Control of the Whole Army, Jiaozuo 454000, *China*

[Abstract] Depression is a kind of mental disorder with high incidence rate and high suicide rate. The pathogenesis has not yet been clarified. At present, the diagnosis is mainly based on the symptoms of patients and the results of mental examination, lacking objective indicators. Functional near-infrared spectroscopy (fNIRS), as a safe and non-invasive brain functional imaging technique, mainly studies depression by measuring hemodynamic changes in the cerebral cortex. Verbal fluency task (VFT), as the most commonly used task paradigm, can effectively evaluate the speech ability and executive function of patients with depression. This article reviews the application of fNIRS combined with VFT in the diagnosis, differentiation, and efficacy evaluation of depressive disorders, and explores its value as a potential biological marker for depressive disorders, which may provide personalized treatment for patients.

[Key words] depressive disorders; functional near-infrared spectroscopy; speech fluency task; diagnosis; efficacy evaluation

抑郁障碍是一种慢性致残性精神类疾病，在中国是心境障碍中发病率最高的疾病，终身患病率和年患病率分别为3.4%和2.1%[1]，被认为是导致残疾的第二大原因[2]。不同患者的临床症状和严重程度差异较大，目前诊断主要依靠病史、精神检查及量表等进行评定，这些方法存在一定的主观性和局限性，容易发生漏诊和误诊[2]。因此，寻找灵敏度及特异度均较高的客观影像学技术是极其重要的。近红外光谱脑功能成像（functional near-infrared spectroscopy，fNIRS）作为新兴的临床检测技术，主要从额颞叶的血流动力学变化及氧合状态变化方面进行抑郁障碍的研究，可用于抑郁障碍的诊断、鉴别诊断、疗效评估及神经机制研究[3]。言语流畅性任务（verbal fluency test，VFT）主要用于评估语言能力、执行功能和认知灵活性，具有操作简单、时间效率高等优点，是研究抑郁障碍应用最多的范式[4]。因此，本文综述基于fNIRS技术对抑郁障碍领域的研究，探究执行VFT时大脑皮质激活情况，分析任务中的血流动力学特征在抑郁障碍诊断、鉴别、疗效评估及神经机制研究等方面的实践价值。

1 fNIRS与VFT

1.1 fNIRS技术原理

fNIRS是一种安全、简单且有效地评估大脑活动的方法[5-6]。该技术主要利用大脑皮层组织中氧合血红蛋白和脱氧血红蛋白对近红外光（600～900 nm）不同波长的吸收率的差异特性，来实时、直接检测大脑皮层的血液活动，进而判断大脑皮层的血流动力学变化[7]。与传统神经影像学技术相比，fNIRS操作简便、成本低、时间分辨率高，具有高生态效度、对运动不敏感、适用于特殊人群（如婴幼儿、多动症患者）及与经颅磁刺激（transcranial magnetic stimulation，rTMS）等技术的良好兼容性，更适用于临床及研究中应用[8]。

1.2 VFT的设计与意义

VFT要求被试者在规定时间内尽可能多地说出符合特定规则的词语，用于评估被试者的语言生成能力、执行功能、注意力及抑制控制等多种认知功能，操作简单，用时短，是神经精神类患者中应用最多的一种认知激活范式[4]。2024年，山西医科大学临床医学学者TONG等[9]在抑郁障碍的神经生物学机制研究中发现VFT可以激活额颞叶皮层，从而使该皮层血流动力学发生变化。2024年，韩国知名学者HUANG等[10]在利用fNIRS技术对抑郁障碍患者进行分类的研究中发现类似的结果，且可以通过fNIRS技术监测VFT期间大脑额颞叶皮层的血流动力学变化情况，为深入了解抑郁障碍的神经病理生理机制提供重要线索，为抑郁障碍的诊断和治疗提供新的思路和方法。

2 fNIRS结合VFT在抑郁障碍中的应用

2.1 诊断与评估

不同类型的抑郁障碍患者，如青少年抑郁障碍、老年抑郁障碍在VFT等认知任务中，大脑激活模式与健康对照组存在差异[11-12]。2024年，华中科技大学教育学院学者DA等[13]在利用fNIRS探讨阈下抑郁的大脑神经活性特征的研究中发现阈下抑郁的右侧背外侧前额叶（dorsolateral prefrontal cortex，DLPFC）和布罗卡区（broca's area，BA）的平均氧合血红蛋白（average oxyhemoglobin level，Avg oxy-Hb）及

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作者简介：范梦影（1990.08-），女，本科，主治医师，研究方向：抑郁障碍
通讯作者：梁学军（1975.09-），男，硕士，副主任医师，硕士生导师，研究方向：常见精神障碍的临床诊疗与生物机制研究

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氧合血红蛋白峰值均低于健康对照组。对于任何年龄段的抑郁障碍患者，fNIRS一致证明，与健康对照组相比，抑郁障碍患者的脑血流动力学变化减弱[14-15]，额叶和双侧颞叶脑区激活下降[16-19]。在VFT范式下对识别抑郁障碍做出重要贡献的通道分别位于DLPFC[11，20-24]、额极区（frontal pole cortex，FPC）[20-22]以及BA[22]，其中抑郁障碍患者DLPFC的氧合血红蛋白（oxyhemoglobin level，oxy-Hb）水平与抑郁障碍的严重程度呈负相关[11，21，23，25-26]。因此，右侧DLPFC和BA的Avg oxy-Hb可能是阈下抑郁的潜在生物学标志物，或可有助于早期识别抑郁。DLPFC的oxy-Hb水平可能有助于区分抑郁障碍的严重程度。

通过搜集受试者在VFT范式下的fNIRS数据，建立判别模型，识别受试者中患有抑郁障碍的个体，其准确率均在75%以上[12，27-28]。2024年，东南大学心理学学者尹营营等[29]在识别抑郁障碍患者准确度的研究中发现，fNIRS诊断抑郁障碍的灵敏度为92%，特异度为78%，其中S积分和峰值对抑郁障碍诊断具有重要价值。

前额叶皮质oxy-Hb可能作为一个客观可测量的诊断参考指标，用于评估和筛查抑郁障碍。其中FPC、DLPFC及BA的oxy-Hb水平对抑郁症状具有较高的相关性，可能作为抑郁症状存在的参考指标，DLPFC的oxy-Hb水平可用作指示抑郁症状严重程度的参考指标。

2.2 鉴别诊断

由于抑郁障碍的临床表现具有高度异质性，且易与其他精神疾病共病，传统的临床诊断往往依赖于主观评估，导致误诊率较高[2]。因此，探索客观、灵敏、可靠的生物标志物，辅助抑郁障碍的鉴别诊断，具有重要的临床意义。将fNIRS与VFT相结合，可以评估抑郁障碍患者在执行认知任务时的脑功能活动模式，从而了解抑郁障碍的神经机制，提高鉴别诊断的准确性。

VFT范式下额叶oxy-Hb的积分值和重心值可以对精神障碍进行分类[30-32]。大量研究发现抑郁障碍患者与焦虑障碍[33-35]、双相情感障碍[30，32，36-37]及精神分裂症[31，38]患者在额叶及颞叶皮层具有不同的血流动力学反应。与焦虑障碍患者相比，抑郁障碍患者在左侧DLPFC[34]、右内侧前额叶皮质[33]、FPC[34]中表现出显著的oxy-Hb低激活及左侧颞上回和中央前回脑区连接网络减少[36]。与双相情感障碍患者相比，单相抑郁患者的额叶积分值更低[30]，通过对比前额叶皮质血流动力学变化的差异，发现抑郁障碍患者左侧DLPFC及腹外侧前额叶皮质（ventrolateral prefrontal cortex，VLPFC）区域血流动力学激活显著较低[36-37]，左侧DLPFC含氧血红蛋白浓度变化与单相抑郁组抑郁症状的严重程度相关[37]。抑郁障碍患者在FPC和眶额叶的oxy-Hb浓度较精神分裂症患者高[38]。

fNIRS检测到的前额叶oxy-Hb浓度及脑区网络连接情况可能用来区分抑郁障碍患者与焦虑障碍、双相情感障碍和精神分裂症患者，将额叶及颞叶的积分值和重心值进行组合，可更好地进行鉴别诊断。

2.3 治疗效果的监测

抑郁障碍目前治疗方法较为多样，治疗结果存在一定的个体差异，多数治疗手段的潜在神经机制不明，且缺乏有效评估治疗反应的生物标志物，识别具有治疗结果预测性的客观生物标志物可以避免无效的治疗方法提高疗效[39]。

2.3.1 评估药物治疗效果：抑郁障碍患者开始药物治疗前的fNIRS信号可能有助于预测抗抑郁药物治疗的临床反应[40-41]。2017年，日本学者MASUDA等[42]在预测抑郁障碍患者对药物应答反应的研究中发现在额叶区域，对抗抑郁药物有应答组治疗前的oxy-Hb水平大于无应答组。2025年，新加坡学者HO等[43]在fNIRS预测抑郁障碍治疗反应的研究中发现DLPFC、左侧颞上回、VLPFC和右内侧前额叶皮质中的总血红蛋白的平均值在应答组和无应答组之间存在显著差异。VFT范式下抑郁障碍患者额下回和颞中回基线Avg oxy-Hb值与治疗后抑郁症状的改善之间存在显著关联[44]。上述研究表明，额颞叶皮质不同的血流动力学变化可能用于估测抑郁障碍患者对抗抑郁药治疗的反应。

2.3.2 评估经颅磁治疗效果：与抗抑郁药物不同，经颅磁刺激（transcranial magnetic stimulation，TMS）通过外部磁场直接调节神经回路，从而影响神经元活性[45]。TMS是治疗抑郁障碍重要且有效的方法[46]。接受重复经颅磁刺激（repetitive transcranial magnetic stimulation，rTMS）治疗的抑郁障碍患者的临床症状改善与额叶血流动力学反应显著相关[47-49]。抑郁障碍患者右侧VLPFC从基线到治疗后oxy-Hb值升高与rTMS后汉密尔顿抑郁量表评分降低呈正相关[50]。2018年，日本学者SHINBA等[48]在脑活动变化与TMS疗效关系的研究中发现TMS在症状明显改善组可观察到额叶血流量的增加与症状改善显著相关，且额叶血流量增加的程度显著高于TMS治疗无效组。因此测定TMS期间额叶oxy-Hb浓度对评估TMS的疗效有一定价值。

2.3.3 评估针灸治疗效果：田乐等[51]发现针灸可以改善脑内神经递质水平，调节抑郁障碍患者大脑皮层的oxy-Hb水平，改善抑郁症状。药物联合针灸治疗后较单用药物治疗抑郁症状改善更明显，且DLPFC脑功能连接更强[52]。2017年，陈窈[53]在观察抑郁障碍患者对针刺响应的研究中发现针刺能增加抑郁障碍患者左侧DLPFC的oxy-Hb水平，改善患者大脑的抑制状态。2021年，中山大学康复医学学者ZHANG等[54]在研究针灸治疗抑郁障碍的相关机制中，发现左侧DLPFC激活程度与抑郁症状的严重程度之间存在相关性。针灸治疗期间通过fNIRS测量左侧DLPFC的oxy-Hb信号可能有助于预测抑郁障碍患者的临床治疗效果。

上述研究表明fNIRS为评估抑郁障碍药物及物理治疗效果提供了一种非常有前景的方法。通过客观测量oxy-Hb或总血红蛋白的变化，可以更好地预测及评估抑郁障碍患者对不同治疗方法的反应，为下一步优化治疗方案，进一步改善抑郁障碍患者的生活质量提供理论支持。

3 总结与展望

综上所述，fNIRS结合VFT作为一种新兴的神经影像学工具，为探索抑郁障碍的诊断、鉴别、疗效评估及神经机制方面提供了客观的量化指标和技术支持。通过fNIRS观察到的前额叶皮质血流动力学变化及脑区功能连接情况可以作为评估抑郁障碍的有效工具，VFT与fNIRS信号检测相结合或许可以进一步揭示抑郁障碍的病理生理机制，有望为患者提供精确的诊断和个体化治疗方案。

然而，fNIRS的空间分辨率低，测量深度有限，但抑郁障碍的功能障碍与大脑的多个区域的异常有关，因此需要将fNIRS与其他脑影像学技术相融合，提高时间和空间分辨率，更加系统、全面、准确地研究抑郁障碍的神经机制。其次，目前多数研究的样本量有限，需要扩大样本量及延长研究周期以降低假阳性和（或）假阴性率，开展长期随访研究，观察抑郁障碍患者脑功能变化的动态过程。基于fNIRS的脑功能特征，开发个性化的治疗方案，提高治疗效果。

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